From Search to Sale: Optimizing Customer Engagement and Product Discovery at JD.com

A Search and Shopping Behavior Analysis

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Tools: Python, Jupyter Notebook, PostgreSQL, pgAdmin, SQL, Tableau, Microsoft Powerpoint, and GitHub
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Executive Summary

The Business Problem

JD.com operates a large e-commerce marketplace where customers must navigate millions of products to find items relevant to their needs. The company needs to understand how customer searches, product impressions, clicks, and previous shopping interactions might affect purchasing behavior. This project analyzes available data to identify patterns in recorded engagement, purchasing activity, demand signals, and segment-level performance. Because the processed data does not link complete customer journeys, the analysis does not identify confirmed search-to-purchase drop-off or causal conversion factors. The findings will be used to recommend improvements to personalized search rankings, targeted marketing, product placement, and product-discovery strategies.

The Dataset, Scope, Tools, and Methods

The analysis uses a processed portion of the JDsearch dataset containing 320,132 recorded positive customer interactions. Python and pandas were used in Jupyter Notebook to inspect, clean, transform, and validate the data. The polished data was then loaded into PostgreSQL, where SQL views were developed to summarize category, product, brand, shop, and query performance. Tableau was then connected to the database to create two interactive dashboards to be used with this report.

The analysis found 223,909 clicks, 75,407 cart additions, and 20,816 purchases. Clicks accounted for 69.94% of recorded positive interactions, cart additions for 23.55%, and purchases for 6.50%. Performance varied across anonymized products, categories, brands, and shops. The opportunity matrices highlighted segments with relatively high recorded engagement and lower Purchase Share. These segments are useful diagnostic priorities, but the current data does not establish that customers encountered a barrier before purchasing.

The Most Notable Results

I noticed that there was a substantial difference between early engagement activity and recorded purchasing activity. In terms of volume, recorded clicks outnumbered purchases by more than ten to one, and cart additions were more than three-and-a-half times the number of purchases. These results show that recorded engagement activity is much more common than recorded purchasing activity in the processed analytical population. The difference is useful for prioritizing further investigation, but it does not show that the same customers progressed from browsing to cart activity and then failed to purchase. Because non-interaction impressions were excluded, the 6.50% Purchase Share should not be interpreted as a traditional conversion rate.

The analysis also revealed meaningful variation across anonymized products, categories, brands, and shops. The opportunity matrices identified segments with comparatively high levels of recorded engagement but below-average Purchase Share. These segments are especially important because they demonstrate existing customer interest without a corresponding level of purchasing activity. They provide strong candidates for further investigation into possible barriers such as pricing, product information, availability, search relevance, shipping terms, or customer trust. At the same time, the highest- performing brands, categories, and shops offer useful benchmarks that JD.com can study and potentially apply to lower-performing segments.

The Business Implications and My Recommendations

The findings have several important business implications for JD.com. The large differences in recorded click, cart-addition, and purchase volumes identify areas where additional diagnostic analysis could be valuable. JD.com should investigate whether product information, pricing competitiveness, inventory availability, delivery transparency, payment convenience, search relevance, or trust signals are associated with lower Purchase Share in high-engagement segments. These are testable business hypotheses rather than causes established by this dataset.

Segments showing substantial engagement and comparatively low Purchase Share should be prioritized for diagnostic review before any intervention is selected. Stronger-performing segments can be examined as comparison groups to generate hypotheses about potentially useful practices. Any proposed change should then be tested with additional operational or linked behavioral data. Because the analysis excludes non-interaction impressions, these findings guide prioritization and investigation rather than directly measure conversion performance or demonstrate potential sales lift.

A Brief Limitation Statement

This analysis is limited by the exclusion of non-interaction impressions, meaning Purchase Share reflects purchases among recorded positive interactions rather than a traditional conversion rate. In addition, product, brand, category, shop, and query identifiers are anonymized, which limits the ability to connect results to specific business entities. The dataset also does not provide sufficient contextual information to determine why customers did not purchase, so the findings identify areas for further investigation rather than establish direct causes.

Final Deliverables

This project's final deliverables include a cleaned dataset containing 320,132 recorded positive customer interactions, a Jupyter Notebook documenting data preparation and analysis, and a PostgreSQL database with reusable SQL views summarizing product, category, brand, shop, and query performance. Two interactive Tableau dashboards present key metrics, purchasing rankings, and engagement opportunity matrices, with filtering and navigation features that support further exploration. The accompanying project report documents methodology, key findings, analytical limitations, and business recommendations, providing both technical documentation and a decision-making resource. Finally, the Microsoft PowerPoint presentation presents all of the findings in a format that is digestible for stakeholders at different levels.

Project Overview

This JD.com Customer Engagement and Purchase Analytics project examines recorded search and shopping behavior to identify patterns, compare purchasing activity, and highlight opportunities for further business investigation. Using the JDsearch dataset, this independent portfolio project explores how customer interaction data can inform product, marketing, and seller-performance decisions.

The analysis focuses on a processed dataset containing 320,132 positive interactions, categorized as clicks, cart additions, or purchases. Performance is examined across anonymized products, categories, brands, shops, and search queries. Particular attention is given to identifying segments with high interaction volumes but relatively low Purchase Share, alongside segments that demonstrate stronger purchasing activity.

The project follows an end-to-end analytical workflow. Python and pandas were used within Jupyter Notebook to inspect, clean, transform, and validate the data. The prepared dataset was then imported into PostgreSQL, where SQL queries and reusable views supported performance comparisons. Tableau was used to develop two interactive dashboards: Executive Overview and Product and Seller Analysis. Together, these dashboards provide access to key metrics, purchasing rankings, and engagement opportunity matrices through filtering and navigation features.

The project's purpose is to translate interaction data into evidence-based recommendations and priorities for further investigation. Because the processed dataset excludes non-interaction impressions, Purchase Share measures purchases as a proportion of recorded positive interactions rather than a traditional conversion rate. Differences between interaction counts also do not establish individual customer journeys or confirmed abandonment. Accordingly, the findings support exploratory decision-making rather than causal conclusions or claims of demonstrated business improvement.

Business Problem and Objective

Business Problem

E-commerce businesses must understand not only which products attract attention, but also where that attention is associated with purchasing activity. Large volumes of search and shopping records can make it difficult to compare performance, identify promising segments, and determine where further investigation would be most valuable.

This independent portfolio project examines that challenge using JD.com's JDsearch dataset. It focuses on differences in recorded engagement and purchasing activity across products, categories, brands, shops, and search queries. Segments with high interaction volumes but relatively low Purchase Share may warrant additional investigation, while stronger-performing segments can provide useful benchmarks. However, these patterns alone do not establish why customers purchased or identify confirmed points of abandonment.

The central business question is:
How can JD.com use recorded search and shopping behavior to identify engagement patterns, compare purchasing performance, and prioritize opportunities for product, marketing, and seller improvement?

Business Problems Addressed

The analysis addresses four related decision-making challenges:

- Understanding the interaction mix: Establishing how recorded activity is distributed among clicks, cart additions, and purchases.
- Comparing segment performance: Identifying differences in interaction volume, purchase counts, and Purchase Share across products, categories, brands, shops, and queries.
- Prioritizing further investigation: Highlighting segments that attract substantial engagement but have comparatively low Purchase Share, while considering the volume of observations supporting each comparison.
- Making findings accessible: Transforming detailed records into reusable summaries and interactive dashboards that help stakeholders explore patterns and evaluate priorities.

Project Objectives

The project was designed to achieve the following objectives:

1. Establish baseline metrics. Calculate total recorded positive interactions, clicks, cart additions, purchases, and Purchase Share using consistent definitions.
2. Prepare reliable analytical data. Inspect and clean the dataset, standardize relevant fields, and validate the information used in subsequent analysis.
3. Compare performance across business segments. Use SQL queries and reusable views to summarize product, category, brand, shop, and query performance.

4. Identify potential improvement opportunities. Examine the relationship between interaction volume and Purchase Share to distinguish high-engagement, lower-purchase segments from stronger-performing benchmarks.
5. Develop interactive decision-support dashboards. Present key metrics, purchasing rankings, and opportunity matrices in Tableau, with filtering and navigation features that support exploration.
6. Produce evidence-based recommendations. Translate observed patterns into practical priorities for further investigation and testing, without claiming that the analysis establishes causes or guarantees improvements.

Scope of Success

Success is defined by accurate and consistent metrics, a documented analytical workflow, reusable SQL summaries, functioning Tableau dashboards, and recommendations traceable to the findings. The project does not measure the impact of implemented interventions or demonstrate increased sales. Because non-interaction impressions are excluded, Purchase Share represents purchases among recorded positive interactions rather than a traditional conversion rate.

Dataset and Scope

Data Source and Analytical Purpose

This project uses data from the JDsearch dataset to examine recorded search-related shopping behavior on JD.com. The analysis brings together query information, product attributes, and interaction labels to compare engagement and purchasing activity across products, categories, brands, shops, and queries.

The project is an independent portfolio analysis rather than an internal JD.com operational assessment. Its findings describe the processed records used in this project and should not be assumed to represent all JD.com customers, transactions, or platform activity.

Dataset Size and Composition

The final analytical dataset contains 320,132 records representing positive interactions, classified as clicks, cart additions, or purchases. The processed data includes 110,393 distinct query record IDs, 237,814 distinct product IDs, and 47,757 distinct brand IDs.

The distribution of interaction labels is summarized below.

Interaction Type	# of Records	Share of Positive Interactions
Click	223,909	69.94%
Add to Cart	75,407	23.55%
Purchase	20,816	6.50%
Total	**320,132**	**100%**

Percentages are rounded to two decimal places, so individual percentages may not sum to exactly 100%.

These figures represent labeled records, not distinct customers. Likewise, distinct query record IDs should not be interpreted as a count of unique people.

Interaction Labels and Inclusion Criteria

The interaction-label mapping used in the project is

Label	Meaning	Treatment in this analysis
0	No Interaction	Excluded from processed dataset
1	Click	Included
2	Add to Cart	Included
3	Purchase	Included

The analytical population therefore consists only of records with positive interaction labels. Here, “positive” means that an interaction occurred; it does not indicate customer satisfaction or a favorable financial outcome.

This restriction supports comparisons within the engaged subset of records but removes the denominator needed for measures based on all displayed candidates or impressions. The report therefore does not calculate an overall click-through rate or an impression-based purchase rate.

Key Fields

The cleaned dataset contains identifiers, query and product information, interaction labels, and derived indicators.

Field	Description	Analytical Use
query	Query content as represented in the dataset	Supports query-level grouping and exploration
query_record_id	Identifier associated with a query record	Supports distinct query-record counts and record tracing
candidate_record_id	Identifier associated with a candidate record	Used in the Tableau record-count measure. It is not treated as a customer identifier
product_id	Anonymized product identifier	Used to compare product performance
product_name_tokens	Tokenized product-name information	Not treated as a reliable, readable product name
brand_id	Anonymized brand identifier	Used to aggregate brand-level activity
category_level_4_id	Anonymized fourth-level category identifier	Used for category comparissons
shop_id	Anonymized shop identifier	Used to compare seller-level activity
interaction_label	Numeric label representing the assigned interaction type	Used to simplify differentiating types of interactions
interaction_type	Readable interaction category derived from interaction label	Used to define interaction_label
added_to_cat_or_purchased	Binary indicator identifying records labeled as either cart additions or purchases	Used to group interaction types for comparisons
Purchased	Binary indicator equal to 1 for purchase-labeled records and 0 otherwise	Used to group interaction types for comparisons

Identifiers are treated as categorical attributes rather than quantities. Their numeric appearance does not imply a meaningful order, distance, or magnitude. For example, a larger brand ID does not represent a larger or more successful brand.

Boundaries and Interpretation

The project does not estimate revenue, profit, average order value, customer lifetime value, or marketing return on investment. These measures require financial and customer-level information beyond the fields used here. It also does not assess seasonality, changes over time, or the effects of a particular campaign, because the processed analytical table does not include a timestamp field used for those purposes.

Anonymization limits the ability to connect findings to recognizable products, brands, categories, or sellers without additional reference information. The analysis can identify a segment for investigation, but it cannot independently explain its pricing, inventory position, delivery conditions, or customer experience.

Finally, the analysis is observational. High engagement combined with low Purchase Share identifies a potential diagnostic priority, not a confirmed operational problem or a proven cause of lost sales. Establishing causes and measuring the benefits of proposed improvements would require additional data, linked behavioral records, and follow-up testing.

Tools and Analytical Process

Analytical Approach

The project followed a structured workflow that moved from data inspection and preparation to database analysis, visualization, and interpretation. Python supported record-level preparation and exploratory analysis, PostgreSQL provided a foundation for reusable analytical queries, and Tableau translated the results into interactive dashboards.

The analysis was descriptive and exploratory. It focused on summarizing recorded activity, comparing performance across segments, and identifying patterns that warrant further investigation. Predictive modeling, causal inference, and measurement of implemented business improvements were outside the project's scope.

Tools and Their Roles

Tool	Role in the Project
Command Line	Assisted in installing neccessary programs, and file and folder creation
Jupyter Notebook	Provided an interactive environment for executing Python code, inspecting outputs, and documenting the preparation and analysis workflow.
Python and pandas	Supported data inspection, transformation, data-type standardization, derived-field creation, aggregation, and validation.
PostgreSQL	Stored the cleaned analytical data and supported SQL-based aggregation and performance comparisons.
PGAdmin	Provided the interface for managing the PostgreSQL database, importing data, running queries, and reviewing results.
SQL	Supported record counts, conditional calculations, grouping, rankings, and reusable performance views.
Tableau	Provided calculated measures, interactive charts, dashboards, filtering, and navigation.
CSV and Tableau Extracts	Supported data transfer between tools and packaging of the dashboard workbook for use without a live local database connection.
Microsoft Word	Assisted in creating and organizing the project report.
Gamma AI	Provided an environment to display the presentation in a way that is digestable for stakeholders at all levels.
GitHub Repository	Assisted in sharing the project and making it easily accessible to potential employers.

The Workflow

1. **Data Inspection and Preparation**

   The workflow began in Jupyter Notebook, where the dataset's structure, column names, data types, interaction labels, and record counts were examined. This inspection established which fields were suitable for grouping, which represented identifiers, and which could support quantitative measures.

Using pandas, the data was prepared for consistent use across Python, PostgreSQL, and Tableau. Relevant identifiers were treated as categorical values rather than numerical measurements. Interaction labels were translated into readable categories, and binary indicators were created to simplify purchase-related calculations. Column-name inconsistencies were resolved so subsequent operations referenced the intended fields.

The prepared dataset was exported for database import. The notebook retained the preparation steps and their outputs, providing a record of how the analytical dataset was produced.

2. **Exploratory Analysis**

   Initial analysis summarized the distribution of clicks, cart additions, and purchases. Counts, percentages, and distinct-identifier summaries established baseline measures against which later SQL and Tableau outputs could be checked. Preliminary bar charts made the interaction mix easier to inspect. Grouped summaries also supported comparisons across products, categories, brands, shops, and queries. This stage helped identify appropriate measures and chart types for the final dashboards.

   The exploration distinguished between activity volume and Purchase Share. A segment with many purchases may also have a large interaction base, while a segment with a high Purchase Share may have relatively few observations. Both measures were therefore considered when evaluating performance.

3. **Database Development and SQL Analysis**

   The cleaned data was imported into the customer_interactions table within the jdcom_analytics PostgreSQL database. pgAdmin was used to manage the database, execute SQL, and inspect query results.

SQL queries summarized the data using record counts, conditional calculations, grouped aggregates, and sorting. Reusable views were developed for:

- Product Performance
- Category Performance
- Brand Performance
- Shop Performance
- Query Performance

These views organized commonly used analytical calculations and reduced the need to rewrite similar queries for each comparison. They also provided a consistent basis for examining interaction volume, purchase counts, and Purchase Share across different grouping levels.

4. **Metric Definition and Consistency**

Metric definitions were maintained across the analytical tools. Recorded purchases were calculated by summing the binary purchased indicator, while clicks and cart additions were counted using their respective interaction labels.

In Tableau, Purchase Share was calculated as:

SUM([Purchased]) / COUNT([Candidate Record Id])

The resulting ratio was formatted as a percentage. Because COUNT excludes null values, this implementation relies on candidate record IDs being populated for the records included in the denominator.

Purchase Share was calculated from the purchase total and interaction total within the current analytical group or filter selection. It was not calculated by averaging the percentages of individual products, brands, categories, or shops. This approach preserves the correct denominator when combining groups.

5. **Tableau Visualization and Dashboard Development**

Tableau was connected to PostgreSQL to build worksheets and dashboards from the prepared data. The visual design used three main components:

- KPI cards to display headline interaction and purchase measures.

- Horizontal bar charts to compare interaction types and rank products, categories, and brands by recorded purchases.

- Scatterplots to examine interaction volume alongside Purchase Share, with bubble size representing recorded purchases.

The opportunity matrices used average reference lines as exploratory benchmarks. These lines helped distinguish comparatively higher- and lower-performing segments but were not treated as statistical significance thresholds or established business targets.

The worksheets were organized into two dashboards. Executive Overview combined headline metrics, the interaction breakdown, category rankings, and the category opportunity matrix. Product and Seller Analysis presented product and brand rankings alongside the shop opportunity matrix.

Interactive filter actions allowed users to select a segment and examine its associated activity in other views. Navigation buttons connected the dashboards, and explanatory notes clarified the meaning and limitations of the displayed measures.

6. **Validation and Functional Testing**

Validation accompanied the workflow rather than being reserved for the final stage. Record counts and interaction summaries served as checkpoints as the data moved between tools. Dashboard KPI values were checked against the established analytical totals.

Functional testing covered chart selections, filter clearing, and navigation between dashboards. Formatting checks addressed readable labels, consistent number formats, equal KPI spacing, and chart sizing. View-specific restrictions, such as top-ten rankings and minimum-interaction filters, were distinguished from the population used for overall metrics.

A Tableau extract and packaged workbook were prepared to support portability and reduce dependence on the local PostgreSQL connection. The extract represents a saved snapshot of the data rather than an automatically updated operational feed.

7. **Interpretation, Reporting, and Presenting**

The final stage translated the analytical outputs into findings and recommendations. High-engagement segments with relatively low Purchase Share were identified as priorities for further investigation, while stronger-performing segments provided potential benchmarks.

Interpretation remained separate from causal explanation. Differences among click, cart-addition, and purchase counts were not treated as confirmed customer drop-off, and the analysis did not establish why a particular segment performed differently. Proposed improvements were therefore framed as areas for investigation and testing, supported by the dashboards and documented in the project report.

Finally, a report was formatted and written, summarizing the entire project. This report serves as the technical documentation. To make the information more digestible for stakeholders at every level, a presentation was also created to simplify the information so that it's more easily understood.

Data Cleaning and Preparation

Preparation Objective

Data cleaning and preparation were performed in Python using pandas within Jupyter Notebook. The objective was to produce a consistently structured dataset suitable for exploratory analysis, PostgreSQL storage, and Tableau visualization. Preparation focused on clarifying field meanings, resolving naming inconsistencies, standardizing data types, creating analytical indicators, and establishing validation totals.

The working analytical DataFrame, clean_df, provided the basis for subsequent summaries and exports. Transformations were documented in the notebook so the preparation logic could be reviewed and repeated.

Initial Data Inspection

The dataset was examined to identify available columns, existing data types, interaction-label values, and the number of records available for analysis. Distinct-identifier counts helped describe the coverage of query records, products, and brands.

This inspection also distinguished identifiers from quantitative measures. Product, brand, category, shop, and record IDs support grouping and record tracing; their numerical appearance does not make them meaningful quantities. Similarly, tokenized product-name information was retained as represented in the source rather than interpreted as ordinary product names.

Column-Name Standardization

Column names were checked for consistency with the names used in analysis code. A naming mismatch involving the category identifier initially caused a KeyError when the preparation code attempted to access category_level_4_id.

Inspecting the actual column list made it possible to identify and resolve the mismatch. The category field was standardized to the name expected by subsequent analysis. Consistent naming reduced the risk of referencing the wrong field and helped align the Python dataset with the PostgreSQL table and Tableau data source.

Column-existence checks were also used during identifier conversion to expose missing or incorrectly named fields. Such checks assist diagnosis but do not replace correcting a required field before downstream analysis.

Data-Type Standardization

Identifier columns were prepared for categorical use, with string conversion applied to ID fields. This prevented identifiers from being treated as values to sum or average and supported consistent grouping and filtering across tools.

The interaction label and binary analytical indicators were represented using integer types. Compact integer storage was appropriate because these fields contain a small range of discrete values. Query content and tokenized product-name information remained text-based.

The resulting distinction between text identifiers and numerical indicators supported reliable calculations. For example, brand_id identifies a group, while purchased provides a value that can be summed to calculate that group's recorded purchases.

Definition of Analytical Population

The processed analytical dataset included records assigned to Click, Add to Cart, or Purchase and excluded records labeled No Interaction.

This exclusion defined the scope of the analysis rather than identifying the excluded records as erroneous. Records without interactions may be useful for other purposes, including impression-based performance measures, but they were outside this project's positive-interaction population.

After preparation, the analytical dataset contained 320,132 records. All overall interaction percentages were calculated using this population. The exclusion of non-interaction records was carried into the report and dashboard methodology notes to prevent Purchase Share from being misinterpreted as an overall conversion rate.

Creation of Derived Fields

Three fields were used to make the interaction labels easier to interpret and analyze.

Derived Field	Preparation Rule	Analytical Purpose
interaction_type	Map label 1 to Click, 2 to Add to Cart, and 3 to Purchase.	Provide readable categories for summaries, filters, and charts.
purchased	Assign 1 when the interaction label is 3; otherwise assign 0.	Calculate purchase counts through summation.
added_to_cart_or_purchased	Assign 1 when the interaction label is 2 or 3; otherwise assign 0.	Identify the combined set of cart-addition and purchase records.

These indicators represent the label assigned to each record. They do not reconstruct a sequence of customer actions. In particular, a purchase-labeled record does not establish that a corresponding click or cart-addition record appears elsewhere in the processed table.

The combined added_to_cart_or_purchased field was kept separate from the standalone Cart Additions measure. Using the combined field as a count of cart additions would incorrectly include purchase-labeled records.

Identifier Validity and Analytical Filters

The shop identifier -1 was treated as a sentinel value rather than a named shop for seller-performance analysis. A Tableau validity filter excluded this value from the shop opportunity view. This was a view-specific restriction, not a blanket deletion from the full analytical dataset.

Similarly, the category opportunity view applied a minimum of 100 recorded interactions to reduce emphasis on very small groups. Top-ten filters restricted the ranking charts to their highest-purchase segments. These restrictions were analytical presentation choices rather than general data-cleaning rules.

Repeated identifiers also require careful interpretation. A product, brand, or query identifier appearing more than once does not, by itself, establish a duplicate record. Identifying exact duplicates requires comparing the relevant record fields, while deciding whether to remove them requires understanding the source data's structure. No specific duplicate-removal or missing-value-imputation totals are reported without a corresponding documented result.

Validation of Prepared Data

Record counts and interaction summaries provided checkpoints for the prepared dataset.

Validation Measure	Prepared-data Result
Total recorded positive interactions	320,132
Click-labeled records	223,909
Add-to-Cart-labeled records	75,407
Purchase-labeled records	20,816
Distinct query record IDs	110,393
Distinct product IDs	237,814
Distinct brand IDs	47,757

The interaction-category counts reconcile to the total:

223,909 + 75,407 + 20,816 = 320,132

The purchase indicator provides a second reconciliation point: its sum should equal 20,816, the number of purchase-labeled records. Likewise, the combined cart-or-purchase indicator should sum to 96,223.

These checks help identify inconsistent label mappings or aggregation logic. They do not, on their own, demonstrate that every field is complete or that the source data is free of errors. In particular, Tableau's use of COUNT(candidate_record_id) requires attention to null identifiers because that calculation excludes null values.

Export and Database Readiness

The prepared dataset was exported to CSV for import into PostgreSQL. The export omitted the pandas DataFrame index so it would not become an unintended database column. Field names, column order, and data types were aligned with the destination table, customer_interactions, in the jdcom_analytics database.

The prepared-data totals served as reference values for checking the database import and subsequent Tableau calculations. This established continuity between the notebook, SQL analysis, and dashboards.

The preparation stage produced a structured analytical dataset while preserving the distinction between source characteristics, cleaning corrections, and view-specific filters. It provided the foundation for comparing engagement and purchasing activity without implying that the records constitute complete customer journeys.

SQL Database and Analysis

Purpose of the Database

PostgreSQL was used to store the prepared dataset and support repeatable analysis through SQL. Moving the cleaned data into a database provided a consistent foundation for calculating metrics, comparing business segments, and connecting the analytical records to Tableau.

The database work focused on descriptive analysis rather than transaction processing. Its purpose was to organize recorded interaction data, produce reusable performance summaries, and support exploration of engagement and purchasing patterns.

Database Structure

The project used a PostgreSQL database named jdcom_analytics, with the prepared records stored in public.customer_interactions. The table retained the query information, anonymized identifiers, interaction labels, and derived indicators described in the Dataset and Scope section.

A single analytical table supported grouping by product, category, brand, shop, or query without requiring joins between separate entity tables. This structure was appropriate for the project's exploratory scope; it was not intended to represent http://JD.com's production database architecture.

Identifiers served as grouping attributes rather than numerical measures. The purchase and interaction indicators supported aggregation. Candidate record IDs were not assumed to be unique row keys, and repeated product or brand identifiers were expected because an entity could appear in multiple records.

Data Import and Validation

The cleaned CSV was imported through pgAdmin into the customer_interactions table. The import required the file's column order, headers, delimiter, and values to align with the destination table.

View	Analytical focus	Purpose
category_performance	Category-level activity	Compare engagement and purchasing patterns across fourth-level category identifiers.
brand_performance	Brand-level activity	Identify brands with higher recorded purchase counts and examine their interaction bases.
shop_performance	Shop-level activity	Support seller comparisons and investigation of engagement–Purchase Share differences.
product_performance	Product-level activity	Support product rankings and identify products for further investigation.
query_performance	Query-related activity	Summarize recorded behavior associated with search queries.

The prepared-data totals provided reference values for validating the import. A basic row-count query established the number of records available in PostgreSQL:

```
SELECT COUNT(*) AS total_rows
FROM public.customer_interactions;
```

The analytical baseline was 320,132 records, comprising 223,909 clicks, 75,407 cart additions, and 20,816 purchases. These totals provided checkpoints for subsequent summaries and Tableau measures.

Validation also required distinguishing total rows from populated identifiers. COUNT(*) counts every row, while COUNT(candidate_record_id) excludes null candidate IDs. Agreement between these counts is important when comparing SQL row totals with the identifier-based count used in Tableau.

Visual Element	Information Presented	Design Rationale
KPI Cards	Headline counts and Purchase Share	Provide an immediate summary of the current analytical population.
Interaction Bar Chart	Counts by interaction type	Make differences in recorded activity easy to compare.
Ranked Horizonal Bars	Purchase counts by product, category, or brand	Support ordered comparisons while accommodating identifier labels.
Opportunity Scatterplots	Interaction volume, Purchase Share, and purchase counts	Reveal combinations of activity and purchasing share that rankings alone may obscure.
Reference Lines	Average values within the configured view	Provide exploratory benchmarks for interpreting segment positions.

Core Analytical Calculations

SQL supported three main types of calculation:

- Counts: Measuring total records, interaction categories, and distinct identifiers.
- Aggregates: Summing purchase indicators and summarizing activity within each segment.
- Ratios: Calculating Purchase Share using the purchase total and interaction total for the same population.

The following illustrative query expresses the project's overall metric definitions:

```
SELECT
  COUNT(*) AS recorded_interactions,
  COUNT(*) FILTER (
    WHERE interaction_label = 1
  ) AS clicks,
  COUNT(*) FILTER (
    WHERE interaction_label = 2
  ) AS cart_additions,
  SUM(purchased) AS recorded_purchases,
  ROUND(
    100.0 * SUM(purchased)
```

```
      / NULLIF(COUNT(*), 0),
      2
   ) AS purchase_share_pct
FROM public.customer_interactions;
```

Multiplication by 100.0 preserves decimal arithmetic and expresses the result as a percentage. NULLIF prevents division by zero when no records satisfy a query's conditions.

Purchase Share was calculated from aggregated counts rather than by averaging segment percentages. This distinction matters because products, brands, and shops have different interaction volumes. A simple average of their percentages would give a small segment the same weight as a much larger one.

Reusable Performance Reviews

Five SQL views were created to organize recurring performance analyses
Views provide named, reusable query definitions. They simplify access to commonly used summaries without requiring analysts to rewrite the underlying aggregation logic each time.

These summaries served as an analytical reference alongside Tableau's calculations. They should not be indiscriminately joined back to the detailed interaction table: doing so can repeat aggregated values across multiple rows and inflate totals. Detailed records and summarized results must be used at their appropriate levels of aggregation.

Segment Comparisons and Rankings

Grouping records by relevant identifiers enabled comparisons of interaction volume, recorded purchases, and Purchase Share. Rankings emphasized recorded purchase counts, while opportunity analysis considered both activity volume and purchasing share.

For example, the following illustrative query ranks categories by recorded purchases:

```
SELECT
   category_level_4_id,
   COUNT(*) AS recorded_interactions,
   SUM(purchased) AS recorded_purchases,
   ROUND(
      100.0 * SUM(purchased)
      / NULLIF(COUNT(*), 0),
      2
   ) AS purchase_share_pct
FROM public.customer_interactions
GROUP BY category_level_4_id
ORDER BY
   recorded_purchases DESC,
   category_level_4_id
LIMIT 10;
```

The secondary ordering field makes the displayed order consistent when purchase counts are tied. A ten-row result is a ranked selection, not a summary of the full category population.

High purchase counts and high Purchase Share answer different questions. Purchase counts describe the volume of purchase-labeled records, while Purchase Share describes their proportion within a segment's positive interactions. Examining both avoids interpreting a large segment as efficient solely because it has many purchases, or treating a small segment's high percentage as sufficient evidence of strong performance.

Analytical Filters and Population Boundaries

Filtering decisions were kept separate from the definition of the full analytical dataset. The category opportunity view used a minimum of 100 recorded interactions, while the shop analysis excluded the sentinel shop identifier -1.

In SQL, an interaction-volume threshold can be applied after aggregation using HAVING COUNT(*) >= 100. A shop validity condition can be applied before aggregation through a WHERE clause. These operations affect the population represented in the output and must be considered when comparing filtered results with overall metrics.

Excluding the sentinel value does not automatically address null shop identifiers, which require separate consideration. Similarly, missing grouping values and repeated identifiers should not be removed without an explicit analytical reason.

Relationship to Tableau

Tableau connected to the PostgreSQL data source to support interactive exploration. The database provided the prepared records and reusable SQL summaries, while Tableau supplied presentation features such as KPI cards, rankings, scatterplots, filters, and dashboard navigation.

Consistent metric definitions were essential across the two environments. Filtered dashboard totals should be compared with SQL results using equivalent conditions, and percentage formatting must be handled consistently: a SQL percentage of 6.50 corresponds to a Tableau ratio of 0.065 displayed as 6.50%.

Analytical Contributions and Limitations

The SQL layer made the analysis repeatable and supported targeted questions about which segments generated activity, which recorded the most purchases, and which combined substantial engagement with relatively low Purchase Share.

These outputs support prioritization, not causal conclusions. The queries do not reconstruct complete customer journeys, establish abandonment rates, or explain purchasing decisions. Their contribution is to identify patterns and provide a consistent basis for further investigation, visualization, and business recommendations.

Key Findings

1. Clicks Dominated the Recorded Interaction Mix

The processed dataset contained 320,132 positive interaction records, with clicks accounting for the largest proportion of activity. Approximately seven out of every ten included records were labeled as clicks. Cart additions represented nearly one-quarter of the records, while purchases represented approximately one in fifteen.

These results establish the composition of the analyzed activity, but they do not describe a linked customer funnel. The differences between counts cannot be interpreted as the number of customers who abandoned a cart or failed to complete a purchase. Establishing those outcomes would require linked customer or session histories.

2. Overall Purchase Share Was 6.50%

The dataset contained 20,816 purchase-labeled records, producing an overall Purchase Share of 6.50% among recorded positive interactions.

This provides a baseline for examining segment-level purchasing patterns. Categories, brands, products, and shops can be compared with that baseline, provided their calculations use the same population definition and relevant filters are considered.

However, Purchase Share is not an impression-based conversion rate. Non-interaction records were excluded, and the denominator therefore represents positive interaction records rather than all visitors, sessions, or displayed products. The metric describes the purchasing composition of the included activity, not the likelihood that an arbitrary JD.com visitor will purchase.

3. Purchases Extended Beyond the Highest-Ranked Categories

The category ranking identified the following leading categories by recorded purchase count:

Together, the ten categories displayed in the ranking accounted for 1,634 purchase records, or approximately 7.85% of the dataset’s 20,816 purchases. The highest-ranked category alone represented approximately 0.89% of recorded purchases.

These results indicate that purchasing activity was not confined to the ten leading categories. Although the ranking identifies useful priorities for closer examination, most recorded purchases occurred outside that selection. Consequently, decisions based only on the top-ten chart would overlook a substantial portion of the analyzed purchasing activity.

The category identifiers are anonymized, so these results cannot be translated into recognizable merchandise categories without additional reference information.

Anonymized Category ID	Recorded Purchases
77255620	185
7929383	177
44610525	173

4. High Interaction Volume Did Not Consistently Coincide With High Purchase Share

The category and shop opportunity matrices showed that segments differed in both interaction volume and Purchase Share. Some segments combined comparatively high interaction volumes with low Purchase Share, while others recorded higher purchasing shares on smaller interaction bases.

This distinction demonstrates why interaction counts alone are insufficient for assessing purchasing performance. A segment can attract considerable recorded activity without purchases representing a similarly large proportion of that activity. Conversely, a high Purchase Share does not necessarily imply a large number of purchases.

High-engagement, low-Purchase-Share segments are therefore candidates for further investigation. Possible explanations include differences in product mix, search intent, pricing, availability, or seller conditions, but the analysis does not establish which explanation applies. These patterns identify where to investigate rather than prove that a particular operational problem exists.

The reference lines displayed in the opportunity matrices provide exploratory benchmarks. They should not be interpreted as statistical significance thresholds, and a chart's average segment percentage is not necessarily equivalent to the overall 6.50% Purchase Share.

5. Broad Product Coverage Requires Caution in Product-Level Interpretation

The processed dataset contained 237,814 distinct product IDs across 320,132 interaction records, which is an average of approximately 1.35 records per product. It also included 47,757 distinct brand IDs and 110,393 distinct query record IDs.

This broad coverage means that product-level observations can be sparse. Although some products have more records than others, a high percentage calculated from a small number of observations may provide limited evidence of sustained performance.

Product rankings therefore describe performance within the analyzed sample, not necessarily long-term demand or platform-wide popularity. Product-level findings are most useful when considered alongside interaction counts, category and brand context, and additional observations where available. The minimum-interaction threshold used in the category opportunity analysis helps reduce small-group effects, but it does not eliminate uncertainty.

Summary of Findings

The analysis established a clear baseline for recorded engagement and purchasing activity and revealed differences across business segments. Clicks dominated the included records, purchases represented 6.50% of positive interactions, and the ten leading categories accounted for only a minority of recorded purchases. The opportunity matrices further demonstrated that interaction volume and Purchase Share capture different aspects of performance.

Collectively, these findings support targeted investigation rather than uniform intervention across all products or sellers. They provide evidence for prioritizing segments, developing hypotheses, and designing follow-up analyses. They do not establish customer abandonment, explain purchasing decisions, or demonstrate that a proposed intervention would increase sales.

Tableau Dashboard Design

Design Purpose

The Tableau dashboards were designed to make the analytical results accessible through a combination of headline metrics, performance rankings, and interactive exploration. The goal was to help viewers understand the overall interaction mix, examine differences across business segments, and identify areas for further investigation.

The presentation was divided into two dashboards: Executive Overview and Product and Seller Analysis. Separating the content reduced the number of competing visuals on each dashboard and allowed users to move from a broad summary to more focused product, brand, and shop analysis.

Executive Overview

The Executive Overview dashboard follows a top-to-bottom structure, beginning with summary metrics and progressing toward detailed comparisons.

Five KPI cards appear across the top:
- Total Interactions
- Clicks
- Cart Additions
- Purchases
- Purchase Share

The cards use prominent numerical values and consistent spacing to support quick interpretation. Their unfiltered values provide the baseline for the analyzed dataset. When a category filter is applied, the cards display the corresponding filtered results rather than the original overall totals.

The middle row contains two horizontal bar charts. Customer Interaction Breakdown compares the counts of clicks, cart additions, and purchases. Top 10 Product Categories by Recorded Purchases ranks anonymized category identifiers by purchase count.

The lower section contains the Category Engagement and Purchase Opportunity Matrix. This scatterplot allows users to examine interaction volume and Purchase Share together, providing context that a purchase ranking alone cannot supply.

Product and Seller Analysis

The second dashboard, titled http://JD.com Product, Brand, and Shop Performance, provides a more focused view of product and seller activity.

Two horizontal bar charts appear side by side across the upper section:
- Top 10 Products by Recorded Purchases
- Top 10 Brands by Recorded Purchases

These charts support comparisons within each entity type. Product and brand totals are interpreted at their respective aggregation levels; a brand may encompass many products, so the two rankings do not represent directly equivalent units.

The full-width lower section contains the Shop Engagement and Purchase Opportunity Matrix. This chart supports investigation of shops with different combinations of interaction volume, recorded purchases, and Purchase Share. The sentinel shop identifier -1 is excluded from this analysis through a validity filter.

Chart Selection and Visual Encoding

Chart types were selected according to the analytical question each view addresses.

The interaction chart uses blue for clicks, orange for cart additions, and green for purchases. Category labels and numerical axes supplement the color encoding so interpretation does not depend on color alone.

In the opportunity matrices, horizontal position represents recorded interactions, vertical position represents Purchase Share, and bubble size represents recorded purchases. Color shading also indicates Purchase Share. These encodings allow viewers to distinguish a segment with substantial purchasing volume from one with a high percentage based on a smaller interaction base.

Reference Lines and Opportunity Interpretation

Average reference lines divide the scatterplots into areas that support exploratory comparison. For example, segments with relatively high interaction volumes but lower Purchase Share can be identified for further investigation, while segments with higher values on both measures may provide useful benchmarks.

These areas are not formal performance classifications. The reference lines do not establish statistical significance, business targets, or the causes of observed differences. Their meaning also depends on the population and filters represented in the view.

The overall Purchase Share is calculated from total purchases divided by total positive interactions. It should not automatically be equated with an unweighted average of individual segment percentages.

Interactive Filtering and Navigation

Filter actions were added to support exploration without requiring viewers to modify worksheets or write queries.

On the Executive Overview dashboard, selecting a category bar filters the associated KPI cards, interaction breakdown, and opportunity view. This allows the user to examine the selected category in greater detail. Clearing the selection restores the broader view, subject to any remaining filters.

On Product and Seller Analysis, the product and brand charts act as filters for the related dashboard views. This supports questions such as which shops are associated with a selected brand or how recorded activity is distributed for a selected product.

Multiple active selections can narrow the population further, so users should clear filters before returning to an overall comparison.

Navigation buttons connect the two dashboards. This creates a straightforward path between the executive summary and the more detailed product and seller views.

Layout and Formatting

Tiled layout containers were used to organize the dashboard components. Horizontal containers aligned the KPI cards and paired bar charts, while the overall vertical arrangement established a consistent reading order.

Formatting emphasized readable titles, clear axis names, and sufficient space for values and labels. KPI cards were distributed evenly, and chart sizing was adjusted to reduce clipping. Axis titles such as Recorded Interactions and Recorded Purchases replaced less accessible technical field names.

The dashboards use a fixed-size desktop layout. Larger views may require scrolling on smaller screens, so the design should not be assumed to provide a fully optimized mobile experience.

Methodology Notes and Responsible Interpretation

Explanatory notes accompany the dashboards to clarify metric definitions and analytical boundaries. In particular, Purchase Share represents purchases among recorded positive interactions rather than a traditional conversion rate.

The charts also preserve anonymized identifiers rather than assigning unsupported product, category, brand, or shop names. Rankings therefore identify records or segments for investigation, not recognizable commercial entities without additional reference data.

The interaction breakdown is presented as a comparison of label counts rather than as a customer funnel. This avoids implying that the dataset tracks every individual through a sequence of clicking, adding to a cart, and purchasing.

Testing and Delivery

Dashboard testing covered filter actions, clearing selections, navigation between dashboards, and the visibility of labels and KPI values. Overall metric displays were checked against the established analytical totals, while filtered views were interpreted according to their selected populations.

A Tableau extract and packaged workbook supported delivery without requiring a live connection to the local PostgreSQL database. The extract represents a saved data snapshot; it does not create an automatically refreshed operational monitoring system.

Together, the dashboards provide an interactive presentation of the project’s findings. Their primary contribution is to make segment comparisons and investigative priorities easier to explore while retaining the context needed to interpret the measures responsibly.

Business Recommendations

Recommendation Framework

JD.com should prioritize investigating and reducing potential friction in the customer journey, particularly within segments that demonstrate substantial recorded engagement but comparatively low Purchase Share. The analysis provides a basis for identifying these segments, but it does not establish why purchasing activity differs or confirm that individual customers abandoned a purchase.

The recommendations below therefore follow a staged approach: identify promising areas for investigation, obtain the additional information needed to understand them, test targeted improvements, and evaluate results before expanding implementation. These are proposed actions, not interventions completed or benefits demonstrated by this project.

1. Prioritize High-Engagement, Low-Purchase-Share Segments

The category and shop opportunity matrices should be used to develop a shortlist of segments for diagnostic review. Segments with meaningful interaction volumes and relatively low Purchase Share are useful starting points because they combine observed engagement with a comparatively small proportion of purchase-labeled records.

Prioritization should consider interaction volume, purchase count, Purchase Share, and the reliability of the underlying observations. A low percentage based on very few records should not receive the same attention as a persistent pattern supported by substantial activity.

Comparisons should also account for differences in product mix and purchasing behavior. For example, expensive or complex products may naturally involve more browsing than routine purchases. A platform-wide average should therefore serve as an initial reference rather than a universal performance target.

The immediate output should be a documented shortlist identifying each segment, its supporting metrics, the questions requiring investigation, and the additional data needed.

2. Investigate Product Information, Pricing, and Availability

For shortlisted segments, JD.com should review the conditions that may influence purchasing decisions. Areas for investigation include:

- Completeness and accuracy of product descriptions and specifications.
- Product-image quality and clarity.
- Pricing relative to comparable alternatives.
- Availability of desired sizes, colors, models, or variants.
- Visibility of shipping charges and delivery estimates.
- Clarity of return, refund, and warranty policies.

- Availability and credibility of customer reviews and seller information.

These factors are diagnostic hypotheses rather than explanations established by the current analysis. Product-page audits, inventory records, pricing information, customer feedback, and service-contact data would be needed to determine which issues are present.

Where a specific problem is identified, improvements should be tested on a limited set of products or shops before broader implementation. Blanket discounting should be avoided unless there is evidence that pricing is the relevant barrier and that the expected benefit justifies the margin impact.

3. Examine Search Relevance and Product Discovery

Query-level summaries provide a foundation for investigating whether displayed products align with search intent. Queries associated with substantial positive interaction activity but limited purchasing activity may warrant closer review.

Further analysis should combine the existing summaries with impression counts, result positions, product attributes, query reformulation, and session information. This would help distinguish between exploratory browsing, irrelevant results, unavailable products, and other possible explanations.

Potential interventions include clearer product titles, improved attribute information, more useful filters, and changes to search-result relevance. These changes should be evaluated against customer outcomes rather than click volume alone. Additional clicks may reflect useful exploration, but they can also indicate that customers are struggling to find a suitable product.

4. Evaluate Cart and Checkout Improvements Using Linked Behavioral Data

The dataset contains more cart-addition records than purchase records, but this difference does not establish a cart-abandonment rate. JD.com should first connect cart events, checkout events, and completed orders using appropriate session or customer identifiers and timestamps.

Once these journeys can be measured reliably, the business could investigate payment errors, unexpected costs, delivery restrictions, required account creation, or other checkout obstacles. Confirmed issues could then inform improvements such as clearer cost disclosure, simplified checkout steps, or better payment-error handling.

Cart reminders or remarketing should be considered only where the business can identify eligible customers, confirm that they have not subsequently purchased, and respect applicable consent and communication preferences. The anonymized analytical dataset used in this project does not support direct customer targeting.

5. Learn From Stronger-Performing Segments Without Overlooking the Wider Assortment

Products, categories, brands, and shops with stronger purchasing performance should be examined as potential benchmarks. Useful comparisons may include product-page completeness, fulfillment conditions, assortment, availability, and customer feedback.

Benchmarking should use reasonably comparable segments. High Purchase Share alone does not establish a superior practice, and high purchase volume may partly reflect greater exposure or a larger assortment.

The analysis also found that the ten leading categories accounted for approximately 7.85% of recorded purchases. This indicates that most purchase activity occurred outside the top-ten selection. JD.com should therefore avoid concentrating all improvement efforts or promotional resources on the most visible rankings.

A balanced approach would investigate established high-volume segments while also considering smaller segments with sufficient evidence of promising performance. Promotional decisions should additionally account for margin, stock availability, service capacity, and incremental demand.

6. Strengthen Measurement and Data Coverage

Additional information is necessary to move from descriptive patterns to more actionable explanations. Priority additions include:

- Non-interaction impressions and search-result positions.
- Event timestamps and privacy-protected session identifiers.
- Linked cart, checkout, and completed-order events.
- Product prices, discounts, and inventory availability.
- Shipping costs, delivery estimates, and fulfillment outcomes.
- Returns, cancellations, and relevant customer feedback.

Authorized reference data would also be needed to connect anonymized identifiers to recognizable products, categories, brands, and shops.

These additions would support properly defined measures such as impression-based click-through rates, session-level purchase rates, and linked cart-completion rates. Metric definitions should specify the population, time window, and unit of analysis so that results remain comparable.

Purchase Share should remain available as a descriptive measure, but it should not be the sole success metric. Its value can change because the mix of recorded interactions changes, even when business outcomes do not improve.

7. Test Interventions and Establish a Review Process

Recommendations should be evaluated through controlled tests where feasible. Each test should define the target population, proposed change, primary outcome, evaluation period, and minimum evidence required before implementation.

Potential outcomes include completed purchases per eligible session and incremental contribution margin, once the necessary data becomes available. Guardrail measures should cover returns, cancellations, customer complaints, and fulfillment performance so that improvements in one metric do not conceal adverse effects elsewhere.

The Tableau dashboards can support periodic review, but the current extract is a static snapshot. Operational use would require a documented refresh process, data-quality checks, and clear responsibility for maintaining the calculations and reviewing results.

Proposed Implementation Priorities

Phase	Priority Actions	Suggested Responsible Functions	Expected Output
Validate and Investigate	Confirm metric definitions, assess data coverage, and shortlist high-engagement segments for review.	Analytics, data engineering, merchandising, and seller operations	A validated baseline and documented diagnostic priorities
Design and Test	Investigate specific barriers and pilot targeted product, search, or checkout improvements.	Product, search, merchandising, marketing, and customer experience	Test results showing whether proposed changes improve relevant outcomes
Scale and Monitor	Expand successful interventions and establish recurring data refreshes and reviews.	Relevant business owners with analytics support	A repeatable process for monitoring benefits and unintended effects

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Expected Business Value

The recommended approach would help JD.com direct investigative and improvement resources toward segments with a clear evidence base. By combining observed engagement patterns with richer operational data and controlled testing, the business could identify practical ways to improve purchasing outcomes and customer experience.

However, no specific sales increase, cost saving, or return on investment is claimed. Those benefits would need to be demonstrated through implementation and evaluation. The principal value of this project is the analytical foundation it provides for making those decisions more systematically.

Limitations

Restricted Analytical Population

The analysis includes only records labeled as Click, Add to Cart, or Purchase. Non-interaction records were excluded from the processed dataset, restricting the findings to recorded positive interactions rather than all product impressions or platform visits.

Accordingly, the overall Purchase Share of 6.50% represents purchases as a proportion of included positive interaction records. It is not a traditional conversion rate, and the available denominator does not support an impression-based click-through or purchase rate. The findings should not be generalized to all http://JD.com activity without evaluating the source dataset's coverage and sampling characteristics.

Incomplete Customer Journeys

The processed table contains query–candidate records with assigned interaction labels rather than a complete chronological history of individual customer behavior. The analysis does not link every click, cart addition, and purchase into a customer or session-level sequence.

Differences between interaction counts therefore do not establish abandonment or progression through a sales funnel. For example, subtracting purchase records from cart-addition records would not produce a valid count of abandoned carts. Similarly, purchase-labeled records should not be interpreted as unique customers, orders, or units sold.

Investigating customer journeys would require appropriately linked event records, timestamps, and session or customer identifiers.

Anonymized Identifiers and Limited Business Context

Product, category, brand, and shop identifiers are anonymized. Although they support grouping and comparison, they cannot be connected to recognizable business entities without authorized reference information. Tokenized product-name information also limits direct interpretation of merchandise characteristics.

The fields used in this project do not provide sufficient information about pricing, discounts, inventory, delivery conditions, reviews, payment problems, or customer motivations. Consequently, the analysis cannot determine which operational factors explain a segment's purchasing performance. Proposed explanations remain hypotheses for further investigation.

Uneven Observation Counts

The dataset contains 237,814 distinct product IDs across 320,132 records, indicating broad product coverage with relatively few observations per product on average. Individual segments vary in size, and percentages based on small groups can change substantially with only a few additional records.

A high Purchase Share supported by limited activity should therefore not be treated as evidence of sustained superior performance. The minimum-interaction threshold used in the category opportunity analysis reduces attention to very small groups but does not establish statistical reliability.

The analysis did not include confidence intervals or formal significance tests. Differences between segments are descriptive and should be evaluated with additional observations before informing consequential decisions.

Aggregation and Filter Effects

Performance varies with the level of aggregation. A brand-level result combines multiple products, while a category or shop may contain a different assortment and mix of purchasing intentions. Differences between these groups may reflect their composition rather than operational effectiveness.

Dashboard filters also change the population represented by the measures. Top-ten rankings display only selected segments, the category opportunity view applies a minimum-interaction threshold, and the shop analysis excludes the sentinel identifier -1. Filtered outputs are not directly interchangeable with overall totals.

Average reference lines provide exploratory context rather than established business targets. An unweighted average of segment percentages may differ from the overall Purchase Share calculated from total purchases and interactions.

Limited Temporal and Financial Analysis

The processed analytical table does not include a timestamp field used to assess trends, seasonality, or changes before and after an intervention. The project therefore describes patterns within the analyzed records rather than demonstrating improvement or deterioration over time.

The analysis also does not establish revenue, profitability, average order value, customer lifetime value, or marketing return on investment. Purchase counts alone cannot determine commercial value because transaction amounts, costs, quantities, and subsequent returns or cancellations were not incorporated into the measures.

Data Quality and Measurement Assumptions

Preparation and validation checks improved structural consistency and enabled reconciliation of key totals across tools. However, agreement between totals does not independently establish that the original records are complete, correctly labeled, or free from collection errors.

Some calculations also depend on field completeness. Tableau's count of candidate record IDs excludes null values, whereas SQL's COUNT(*) includes all rows. These measures are equivalent only when the relevant identifiers are populated. Repeated identifiers likewise require contextual interpretation and do not automatically indicate duplicate records.

Observational Findings and Unverified Business Impact

This project is descriptive and observational. High engagement combined with low Purchase Share identifies a potential investigative priority, but it does not prove that customers encountered friction or establish a specific cause.

No recommended intervention was evaluated through a controlled experiment as part of this analysis. Claims of increased sales, reduced abandonment, or improved profitability would therefore be premature. Those outcomes would require additional data, defined success measures, and follow-up testing.

The Tableau extract is also a saved snapshot rather than an automatically refreshed operational feed. Ongoing business use would require a maintained refresh process and recurring validation.

Implications for Use

The findings are best used to prioritize questions, identify segments for review, and guide the collection of additional evidence. They provide a structured analytical foundation, but business decisions should combine these results with operational context, appropriate comparison groups, and testing before recommendations are implemented at scale.

Conclusion

The JD.com Customer Engagement and Purchase Analytics project demonstrated how an end-to-end analytical workflow can transform search-related shopping records into structured business insights. Using Python, PostgreSQL, SQL, and Tableau, the project produced a prepared dataset, reusable performance summaries, and two interactive dashboards that support exploration across products, categories, brands, shops, and queries.

The analysis examined 320,132 recorded positive interactions, comprising 223,909 clicks, 75,407 cart additions, and 20,816 purchases. Purchases represented 6.50% of the included records. Segment-level comparisons showed that interaction volume and Purchase Share capture different aspects of performance: substantial engagement did not consistently coincide with a high purchasing share. These patterns provide a basis for identifying segments that warrant closer investigation and stronger-performing groups that may offer useful benchmarks.

The principal recommendation is to investigate potential purchasing barriers within high-engagement, low-Purchase-Share segments. Product information, pricing, availability, search relevance, delivery transparency, and checkout conditions are appropriate areas for diagnostic review. However, the current data does not establish which factors explain observed differences. Additional operational information and linked behavioral records should guide targeted interventions, with controlled testing used to evaluate their effects before broader implementation.

The findings must also be interpreted within the project's scope. Non-interaction impressions were excluded, so Purchase Share is not a traditional conversion rate. The records do not establish complete customer journeys, confirmed abandonment, or causal relationships. Anonymized identifiers and uneven observation counts further limit the conclusions that can be drawn about individual business entities.

Ultimately, the project's contribution is a documented, reusable foundation for evidence-based investigation rather than proof of increased sales or improved conversion. It connects data preparation, database analysis, visualization, and business reasoning in a way that makes performance patterns accessible while preserving their limitations. With richer data and follow-up testing, this foundation could support more informed product, marketing, and seller-management decisions.

Appendix

This appendix provides supporting definitions, calculation logic, database references, and reproduction instructions. SQL examples are supplementary reference queries; they should not be interpreted as an execution log or evidence that every check has passed.

Appendix A: Source and Project Materials

Dataset Reference

- Dataset: JDsearch
- Source URL or repository: https://www.kaggle.com/datasets/duuuscha/jd-search-dataset
- Dataset publication or documentation citation: Duuuscha. (n.d.). *JD.com search dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/duuuscha/jd-search-dataset
- Downloaded file or dataset version: archive.zip
- Date accessed: 09/18/2026
- Usage and redistribution terms: You may use, copy, analyze, transform, and build upon the dataset. You must give appropriate credit to the dataset/source when using or redistributing it. You may not use the dataset for commercial purposes. You may redistribute the original data or adaptations, provided you follow the attribution and noncommercial requirements. If you modify or adapt the dataset and distribute that adaptation, you must distribute it under **CC BY-NC-SA 4.0**, the same license as the original.

The analysis uses a processed subset containing positive interaction records. It should not be described as covering the complete source dataset or all JD.com activity.

Supporting Materials

Material	Purpose	File or Location
Jupyter Notebook	Documents preparation, transformation, and exploratory analysis.	JDcom_Shopping_Behavior_Analysis.ipynb
Cleaned Dataset	Supplies the records used for database analysis and visualization.	jdsearch_cleaned_interactions.csv
SQL Scripts	Document table definitions, views, and analytical queries.	01_data_validation_and_setup.sql, 02_category_performance_analysis.sql, 03_brand_and_shop_performance.sql, 04_product_and_query_performance.sql, 05_data_quality_validation.sql
Tableau Packaged Workbook	Contains the dashboards and associated extract.	JDcom_Shopping_Behavior_Analysis.twbx
Published Visualization	Provides browser-based access to the dashboards.	[Insert Tableau Public URL, if published]
Project Report	Explains the methodology, findings, recommendations, and limitations.	JDcom_project_report.docx
Powerpoint Presentation	Summarizes the project report in a less technical and more understandable format	

Source data should be redistributed only where permitted by its usage terms. Database credentials and other sensitive connection information should not be included in shared project materials.

Appendix B: Data Dictionary

The following descriptions refer to the cleaned analytical fields. Identifier roles do not imply that a field is unique or subject to a database primary-key constraint.

Field	Logical Role	Definition
Query	Text	Query content as represented in the dataset.
query_record_id	Identifier	Identifies a query record; not a customer identifier.
candidate_record_id	Identifier	Candidate reference used in record-count calculations; not assumed to be a unique row key.
product_id	Identifier	Anonymized product reference.
product_name_tokens	Text	Tokenized product-name information.
brand_id	Identifier	Anonymized brand reference.
category_level_4_id	Identifier	Anonymized fourth-level category reference.
shop_id	Identifier	Anonymized shop reference; -1 is excluded from the shop opportunity view.
interaction_label	Integer category	Assigned interaction label.
interaction_type	Text category	Readable interaction description.
purchased	Binary indicator	Equals 1 for purchase-labeled records and 0 otherwise.
added_to_cart_or_purchased	Binary indicator	Equals 1 for cart-addition or purchase labels and 0 otherwise.

Interaction Mapping

Label	Description	Analytical treatment
0	No interaction	Outside the processed analytical population
1	Click	Included
2	Add to Cart	Included
3	Purchase	Included

These labels classify records. They do not establish a chronological sequence of actions for an individual customer.

Appendix C: Metric Definitions and Reference Totals

Metric	Definition	Unfiltered reference value
Recorded Interactions	Number of included records	320,132
Clicks	Records with interaction label 1	223,909
Cart Additions	Records with interaction label 2	75,407
Recorded Purchases	Sum of purchased	20,816
Purchase Share	Purchases divided by positive interaction records	6.50%
Distinct Query Records	Distinct query_record_id values	110,393
Distinct Products	Distinct product_id values	237,814
Distinct Brands	Distinct brand_id values	47,757

The interaction totals reconcile as follows:

223,909 + 75,407 + 20,816 = 320,132

The expected total for the combined cart-or-purchase indicator is:

75,407 + 20,816 = 96,223

Purchase Share is calculated within the applicable population or filter selection:

Purchase Share = Recorded Purchases ÷ Recorded Positive Interactions

The ratio is multiplied by 100 for percentage reporting. It is not a visitor, session, or impression-based conversion rate.

Appendix D: PostgreSQL Reference

Database Objects

- Database: jdcom_analytics
- Base table: public.customer_interactions

The project includes the following performance views:

View	Analytical focus
category_performance	Category-level performance
brand_performance	Brand-level performance
shop_performance	Shop-level performance
product_performance	Product-level performance
query_performance	Query-related performance

The saved SQL scripts should contain the exact definitions of these objects. The queries below illustrate validation and calculation logic without creating or modifying database objects.

D1. Record and Indicator Checks SQL

```
SELECT
  COUNT(*) AS total_rows,
  COUNT(candidate_record_id) AS populated_candidate_ids,
  COUNT(*) FILTER (
    WHERE interaction_label IS NULL
       OR interaction_label NOT IN (1, 2, 3)
  ) AS unexpected_label_rows,
  COUNT(*) FILTER (
    WHERE purchased IS DISTINCT FROM
      CASE
        WHEN interaction_label = 3 THEN 1
        ELSE 0
      END
  ) AS purchase_indicator_mismatches,
  COUNT(*) FILTER (
    WHERE added_to_cart_or_purchased IS DISTINCT FROM
      CASE
        WHEN interaction_label IN (2, 3) THEN 1
        ELSE 0
      END
  ) AS combined_indicator_mismatches
FROM public.customer_interactions;
```

For the defined analytical population, unexpected labels and indicator mismatches should be investigated. Agreement between total rows and populated candidate IDs is required for SQL row counts and Tableau’s candidate-ID counts to represent the same denominator.

D2. Interaction Distribution SQL

```
SELECT
  interaction_label,
  interaction_type,
  COUNT(*) AS recorded_interactions,
  ROUND(
    100.0 * COUNT(*)
    / SUM(COUNT(*)) OVER (),
    2
  ) AS interaction_share_pct
FROM public.customer_interactions
GROUP BY
  interaction_label,
  interaction_type
ORDER BY interaction_label;
```

D3. Overall Purchase Share SQL

```
SELECT
  COUNT(*) AS recorded_interactions,
  SUM(purchased) AS recorded_purchases,
  ROUND(
    100.0 * SUM(purchased)
    / NULLIF(COUNT(*), 0),
    2
  ) AS purchase_share_pct
FROM public.customer_interactions;
```

NULLIF protects against division by zero. Decimal arithmetic preserves the fractional result.

Appendix E: Tableau Calculations and Settings

Core Calculations

Purchase Share

```
SUM([Purchased]) / COUNT([Candidate Record Id])
```

The result is formatted as a percentage.

Interaction Count

```
1
```

Summing this row-level calculated field counts the records represented in the view.

Valid Shop

```
[Shop Id] <> “-1"
```

The shop opportunity view retains records where this condition is true. Null shop IDs require separate consideration because comparisons involving null values do not return true.

Dashboard Configuration

Component	Configuration
Executive Overview	Five KPI cards, interaction breakdown, top categories, and category opportunity matrix
Product and Seller Analysis	Top products, top brands, and shop opportunity matrix
Category opportunity filter	Minimum of 100 recorded interactions
Shop opportunity filter	Exclude sentinel shop ID -1
Ranking measure	Recorded purchases
Scatterplot horizontal position	Recorded interactions
Scatterplot vertical position	Purchase Share
Scatterplot size	Recorded purchases
Scatterplot color	Purchase Share
Interactive actions	Category, product, and brand selections filter related dashboard views
Navigation	Buttons connect the two dashboards

Reference-line settings should be preserved in the workbook. An average of segment percentages is not necessarily the same as the overall purchase-to-interaction ratio.

Appendix F: Dashboard Figures

Figure F1. Executive Overview

[Insert an exported image of the complete, unfiltered Executive Overview dashboard.]

Caption: Overview of recorded positive interactions, purchasing activity, leading categories, and category-level engagement opportunities.

Figure F2. Product and Seller Analysis

[Insert an exported image of the complete, unfiltered Product and Seller Analysis dashboard.]

Caption: Product and brand purchase rankings with the shop engagement and Purchase Share opportunity matrix.

Figure F3. Filtered Exploration Example — Optional

[Insert a screenshot showing a selected category, product, or brand.]

Caption: Example of dashboard filtering. Identify the selected segment and any additional active filters so the displayed population is clear.

Appendix G: Reproduction and Review Procedure

1. Obtain the documented source files under the applicable usage terms.
2. Open the preparation notebook and run its cells in order, using the correct local file paths.
3. Review the resulting schema, interaction labels, indicators, and reference totals.
4. Export the cleaned dataset without the pandas index.
5. Create the PostgreSQL objects using the saved project SQL scripts.
6. Import the cleaned data once into the intended table, avoiding duplicate loads.
7. Run validation queries and investigate differences from the documented reference totals.
8. Connect Tableau to the intended database table or open the packaged workbook.
9. Confirm that calculated fields and view-specific filters match the documented definitions.
10. Clear interactive selections before checking the overall KPI values.
11. Test filter clearing and navigation between dashboards.
12. Refresh the extract when appropriate and save a separately identifiable workbook version.

Environment Record

Component	Version or detail
Python	[Insert version]
pandas	[Insert version]
Jupyter Notebook or JupyterLab	[Insert version]
PostgreSQL	[Insert version]
pgAdmin	[Insert version]
Tableau	[Insert version]
Final extract creation date	[Insert version]
Workbook/report revision	Phase 2 final editorial revision

These details help distinguish software differences, data revisions, and filter changes from genuine discrepancies in analytical results.